

iii) TF - IDF:

(Term Frequency - Inverse Document Frequency)

* After lowering the cases & removing stopwords

* we get.

1: good boy

2: good girl

3: boy girl good

$$\Rightarrow \text{Term Freq (TF)} = \frac{\text{Nbr of Repeatability of word in Sentence}}{\text{Nbr of words in Sentence.}}$$

$$\Rightarrow \text{IDF} = \log_e \left(\frac{\text{Nbr of Sentences}}{\text{Nbr of Sentences containing the words}} \right)$$

↳ Term Frequency:

Vocabulary

• good

• boy

• girl

S₁

1/2

S₂

1/2

S₃

1/3

1/2

0/2

1/3

0/2

1/2

1/3

• in sentence 1 $\Rightarrow$ how many times the word "good" repeats?

$\rightarrow$ once

• Total words in sentence 1 are "two"

So $TF = \frac{1}{2}$

↳ IDF

Total number of sentences = 3
number of sentences containing "good" = 3

Vocabulary

• good

• boy

• girl

IDF

$$\log_e (3/3)$$

$$\log_e (3/2)$$

$$\log_e (3/2)$$

*** Final TF - IDF

Multiply TF with ID for each word in sentence

good

boy

girl

$$S_1: \quad \frac{1}{2} \times \log_e (3/3)$$

$$\frac{1}{2} \times \log_e (3/2)$$

$$\frac{1}{2} \times \log_e (3/2)$$

$$S_2: \quad \frac{1}{2} \times \log_e (3/3)$$

$$\frac{1}{2} \times \log_e (3/2)$$

$$\frac{1}{2} \times \log_e (3/2)$$

$$S_3: \quad \frac{1}{3} \times \log_e (3/3)$$

$$\frac{1}{3} \times \log_e (3/2)$$

$$\frac{1}{2} \times \log_e (3/2)$$

Advantages

- Intuitive
- fixed size
- words importance is getting captured $\downarrow$

- if a word is repeated many times in all the sentences which means that word is not that imp.

Disadvantages

- sparse Matrix still exists.